How Does Conformal Coverage Degrade Under Distribution Shift? A Controlled Study of Image Corruption and Label Shift on CIFAR-10

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Problem Description

Conformal prediction is a distribution-free framework for converting the output of a trained classifier into a prediction set rather than a single label. Using a held-out calibration set, a conformal predictor can target marginal coverage of at least $1 - \alpha$ under exchangeability between calibration and test examples. For example, at $1 - \alpha = 0.9$, the long-run proportion of prediction sets containing the true label is controlled near 90% under the standard assumptions. The guarantee is therefore distribution-free but not assumption-free: when the test population differs from the calibration population, the usual finite-sample guarantee no longer automatically applies, and empirical coverage may fall below the nominal target.

Our original design focused on common image corruptions such as noise, blur, weather effects, and digital distortions. These conditions are useful because they create controlled changes in image quality, but they represent closely related forms of input shift. Recent work on broad out-of-distribution evaluation emphasizes that methods may behave very differently across qualitatively different shift mechanisms, even when they perform strongly on one particular benchmark. To keep the project manageable while making the distribution-shift question broader, we study two distinct forms of shift: *image corruption* and *label shift*.

For image corruption, we calibrate on clean CIFAR-10 and evaluate on representative CIFAR-10-C corruptions at several severity levels. For label shift, we keep the clean class-conditional image distributions fixed but change the proportions of the ten CIFAR-10 classes in the test population. In both cases the label space remains unchanged, so empirical conformal coverage remains directly interpretable.

We compare three conformal scoring methods—LAC, APS, and RAPS—using two ResNet-18 base classifiers: a standard model and a model trained with AugMix. The architecture is held fixed so that the effect of robustness-oriented training can be separated from architectural differences. Our goal is not to propose a new conformal guarantee under distribution shift, but to characterize how conformal reliability and prediction-set efficiency depend on the shift mechanism, shift magnitude, conformal score, and robustness of the underlying classifier.

Research Questions

RQ1. How do representative image corruptions and label shift affect empirical coverage and prediction-set size for conformal predictors calibrated on clean CIFAR-10?

RQ2. Do LAC, APS, and RAPS differ in their sensitivity to these two forms of distribution shift?

RQ3. Does robustness-oriented training of the base classifier reduce conformal coverage degradation under image corruption, and does any benefit extend to label shift?

Related Works

Conformal prediction provides distribution-free prediction sets with finite-sample marginal coverage under exchangeability. Given a trained predictive model and a held-out calibration set, conformal methods compute a score for each calibration example and use an empirical quantile of these scores to construct prediction sets for future observations [1]. The framework does not require a parametric model for the data distribution, but its standard coverage guarantee relies on calibration and test examples being exchangeable.

The behavior of conformal prediction when this assumption is violated has also been studied. Tibshirani et al. consider conformal prediction under covariate shift, where the covariate distribution changes between calibration and testing while the conditional response distribution is assumed to remain unchanged [6]. They show that likelihood-ratio weighting can modify conformal inference to recover coverage under appropriate covariate-shift assumptions. Their work establishes that distribution shift may require explicit changes to the usual calibration procedure. Our project instead studies the empirical behavior of standard conformal predictors after the calibration and test populations differ.

For multiclass classification, different conformal scores can produce prediction sets with different efficiency characteristics. A common baseline is the Least Ambiguous Classifier (LAC) score, $s(x, y) = 1 - \hat{p}_y(x)$, which produces prediction sets by thresholding individual class probabilities [1]. Romano et al. introduced Adaptive Prediction Sets (APS), which use cumulative predicted probability mass after ranking classes from most to least likely [5]. Angelopoulos et al. subsequently proposed Regularized Adaptive Prediction Sets (RAPS), which add a penalty for lower-ranked classes to reduce unnecessarily large prediction sets while retaining the standard marginal coverage guarantee under exchangeability [2].

A separate body of work studies robustness to changes in the input distribution. Hendrycks and Dietterich introduced CIFAR-10-C and related corruption benchmarks, which apply common perturbations at multiple severity levels while preserving the original class labels [3]. AugMix is a robustness-oriented data augmentation method that improves performance and uncertainty quality under common image corruptions without requiring exposure to CIFAR-10-C during training [4]. More recent work argues for evaluating reliability across qualitatively different forms of distribution shift rather than many variants of a single shift family. Guille-Escuret et al. benchmark multiple OOD detection methods across novel classes, adversarial perturbations, synthetic images, corruptions, and multi-label inputs, and show that strong performance on one shift type does not consistently transfer to others [7]. This motivates our decision to complement controlled image corruptions with a distinct label-shift experiment while keeping the overall scope appropriate for a course project.

Proposed Work

Data and clean split. We use CIFAR-10 as the in-distribution dataset. The 50,000-image training set is used only to train the base classifiers. The 10,000-image CIFAR-10 test set is divided into disjoint calibration and evaluation subsets. For each split, 5,000 clean images are used for conformal calibration and the remaining 5,000 images are used for all clean and shifted evaluations. Because CIFAR-10-C is generated from the CIFAR-10 test images, shifted evaluation uses only the corrupted counterparts of the evaluation indices; corrupted versions of calibration images are never included in the test set. We repeat the calibration/evaluation split three times and report mean and standard deviation across splits.

Image-corruption shift. To avoid a large number of closely related corruption conditions, we select three representative CIFAR-10-C corruptions: Gaussian noise, defocus blur, and brightness. Each is evaluated at severity levels 1, 3, and 5. Together with the clean baseline, this gives ten image conditions. No recalibration is performed after corruption: all conformal thresholds are estimated on clean calibration data and then kept fixed.

Label shift. We separately study changes in the test label distribution using only clean evaluation images. Starting from the approximately balanced CIFAR-10 class distribution, we define mild, moderate, and severe imbalance scenarios by increasing the probability mass assigned to a small set of majority classes and decreasing the mass of the remaining classes. To avoid conclusions being driven by one specific class, we rotate the identity of the majority classes and average results across these choices. Since the prediction set for each image is already available, label-shift results are computed by reweighting class-conditional quantities; this requires no model retraining, additional neural-network inference, or conformal recalibration.

Base classifiers. We train two ResNet-18 classifiers on the CIFAR-10 training set. The first uses standard random-crop and horizontal-flip augmentation. The second uses AugMix as a robustness-oriented training procedure [4]. Both models use the same architecture and do not see CIFAR-10-C images during training. Model outputs for clean and corrupted evaluation images are cached once and reused by all conformal methods.

Conformal methods. For each base model, we construct prediction sets at target coverage $1 - \alpha = 0.9$ using three score functions. LAC uses $s(x, y) = 1 - \hat{p}_y(x)$. APS uses the cumulative probability mass of classes ranked above and including y, with the standard randomization used for adaptive prediction sets [5]. RAPS extends APS with a regularization penalty on lower-ranked classes [2]. RAPS regularization settings are fixed in advance from standard literature-guided choices rather than tuned on shifted test data. Each model–score pair therefore defines one clean-calibrated conformal predictor whose parameters remain fixed across all evaluation conditions.

Experimental design. The experiment contains six model–conformal-method combinations (2 base models $\times$ 3 conformal scores). For corruption shift, each combination is evaluated on the clean baseline and on the nine corruption/severity conditions. For label shift, the same clean prediction sets are reweighted under several target class distributions. The two shift mechanisms are analyzed separately rather than crossed with each other, preventing a combinatorial expansion of conditions while still allowing comparison across qualitatively different forms of shift.

Evaluation Metrics

The primary metrics are empirical coverage and average prediction-set size. Empirical coverage is

$$\widehat{C} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{y_i \in C(x_i)\},$$

and is compared with the nominal target of 0.9. We report the coverage gap

$$0.9 - \widehat{C}$$

to quantify empirical undercoverage after distribution shift. We use confidence intervals or split-to-split variability rather than treating a fixed numerical gap as a formal "violation" threshold, since the standard exchangeability guarantee no longer applies after the test distribution changes.

Average prediction-set size is

$$\frac{1}{N} \sum_{i=1}^{N} |C(x_i)|,$$

which measures efficiency. Coverage and set size are interpreted jointly because high coverage obtained only by returning many of the ten labels has limited practical value. We additionally report top-1 classification accuracy under each image condition. For the label-shift study, we report class-conditional coverage and class-conditional average set size so that changes in marginal coverage can be traced to changes in class proportions rather than to changes in the prediction set of an individual image.

The main visualizations will be (i) coverage and prediction-set size versus corruption severity for each corruption type, method, and base model, and (ii) marginal coverage versus label-shift severity together with the underlying class-conditional coverage profile. These plots directly address whether conformal reliability deteriorates similarly across the two shift mechanisms and whether robustness-oriented training changes that behavior.

Reference

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